

GridPulse Milestone 3 Written Analysis

Team 9, Intro to Cloud Computing

1. Image size and how we reduced it

Our final backend image is 192 MB compressed in Artifact Registry (804 MB on disk). Built with our Milestone 1 Dockerfile it was 821 MB, so it is now about 77% smaller. The frontend image is 21.2 MB compressed. The biggest saving came from installing `xgboost-cpu` instead of `xgboost`, because the regular package pulls in CUDA libraries and `nvidia-nccl-cu12`, which we never use. We also used multi-stage builds, so the `pip` build stage, the Node toolchain and `node_modules` never reach the final images and the frontend runs on `nginx:alpine` with only the built files. Last, we used `python:3.11-slim`, turned off the `pip` cache and deleted the test folders that ship inside `pandas`, `numpy`, `scipy` and `scikit-learn`.

2. What `.dockerignore` excludes and why

Our `.dockerignore` keeps `.env` files, keys and credential files out of the build context, so they can never be copied into a layer. It also excludes `node_modules`, `venv` and `__pycache__`, which were built on our Macs and would overwrite the Linux packages installed in the image. We also left out `.git` (full commit history), `uploads/` (runtime data that belongs in a volume) and things not needed at runtime like `frontend/dist`, `docs/`, `database/`, `backend/tests` and `.tar.gz` archives. The build context is only 1.2 MB.

3. No secrets baked into the images

Our only secret is the Postgres password. It lives in a `.env` file that both `.gitignore` and `.dockerignore` exclude, and `docker compose` passes it to the containers when they start, so it is never inside an image. Our script `deploy/verify_no_secrets.sh` checks the environment variables in each image and every `docker` history entry, looks for `.env` files and scans every layer exported with `docker save`. Both images passed with 0 matches, and when we tested it on an image with a password baked in on purpose, it caught all four problems.

4. Cost control

We set a \$50 monthly budget on project `gridpulse-509202` with email alerts at 10, 20, 25, 50 and 100%, based on cost before free-trial credits. October spend so far is \$2.42, all covered by credits. The registry and the VM are both in `us-central1`, so pulling images has no egress cost, and a cleanup policy deletes untagged images after 7 days and always keeps the 5 newest versions.